

ELASTIC BEANSTALK

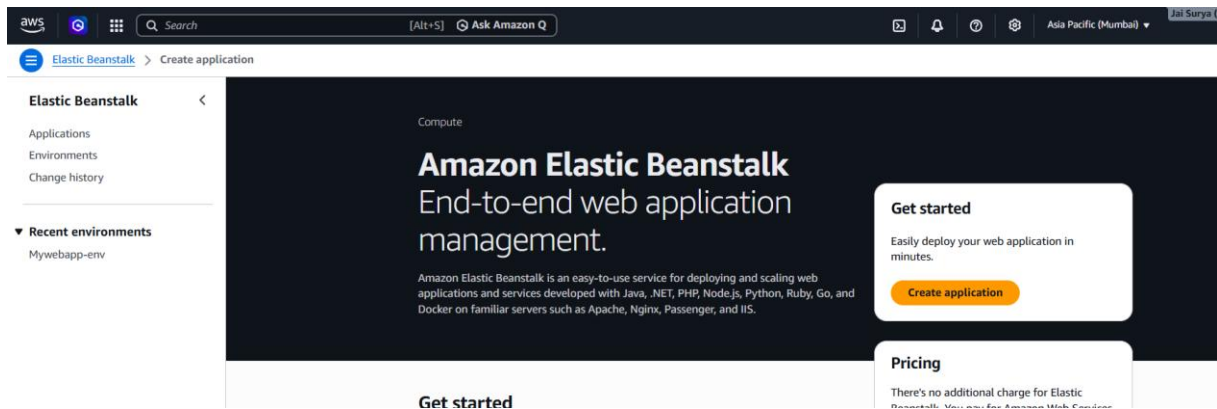
Elastic Beanstalk is a PaaS service that lets us deploy applications easily. It automatically manages infrastructure like EC2, Auto Scaling, and Load Balancer, so we can focus only on the application.

It allows you to deploy and manage applications without handling infrastructure.

HANDS-ON FOR ELASTIC BEANSTALK

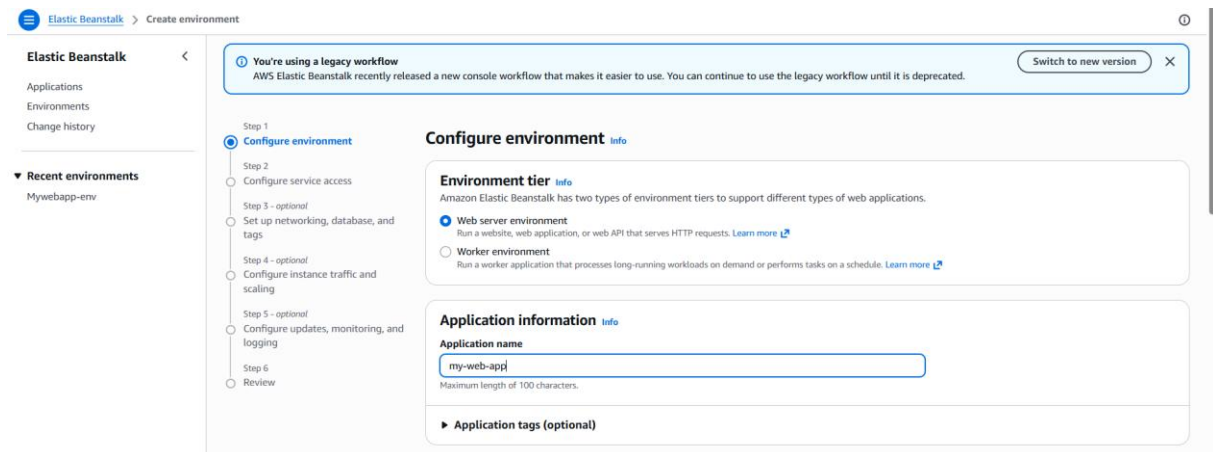
Screenshot: Create Application

- Go to Amazon Web Services
- Click Elastic Beanstalk
- Click Create application



Screenshot: Configure Environment

- Choose web server environment
- Enter application name



Screenshot: Select Platform

- Choose platform as Python
- Select sample application
- Click Next

The screenshot shows the AWS Elastic Beanstalk 'Create environment' console. The left sidebar contains 'Elastic Beanstalk' with links to 'Applications', 'Environments', and 'Change history'. Below this is a section for 'Recent environments' showing 'Mywebapp-env'. The main content area is titled 'Create environment' and contains three sections: 'Platform', 'Application code', and 'Presets'. The 'Platform' section has three dropdown menus: 'Platform' (set to 'Python'), 'Platform branch' (set to 'Python 3.14 running on 64bit Amazon Linux 2023'), and 'Platform version' (set to '4.12.0 (Recommended)'). The 'Application code' section has two radio buttons: 'Sample application' (selected) and 'Upload your code'. The 'Presets' section has a heading 'Configuration presets' and four radio buttons: 'Single instance (free tier eligible)' (selected), 'Single instance (using spot instance)', 'High availability', and 'High availability (using spot and on-demand instances)'. At the bottom, there is a 'Custom configuration' option.

Screenshot: Configure Service Role

- In service role, click Create role

The screenshot shows the AWS Elastic Beanstalk 'Configure service access' console. The left sidebar is the same as the previous screenshot. The main content area is titled 'Configure service access' and contains a 'Service access' section. This section has three sub-sections: 'Service role', 'EC2 instance profile', and 'EC2 key pair - optional'. Each sub-section has a dropdown menu and a 'Create role' button. The 'Service role' dropdown is set to 'aws-elasticbeanstalk-service-role'. The 'EC2 instance profile' dropdown is set to 'aws-elasticbeanstalk-ec2-role'. The 'EC2 key pair - optional' dropdown is set to 'Choose a key pair'. At the bottom, there are buttons for 'Cancel', 'Skip to Review', 'Previous', and 'Next'. A notification banner at the top of the main content area states: 'You're using a legacy workflow. AWS Elastic Beanstalk recently released a new console workflow that makes it easier to use. You can continue to use the legacy workflow until it is deprecated. Switch to new version'.

Screenshot: Create Service Role

- Select AWS service
- Choose use case as Elastic Beanstalk
- Select Elastic Beanstalk environment

Step 1: Select trusted entity

Step 2: Add permissions

Step 3: Name, review, and create

Select trusted entity info

Trusted entity type

- ☒ **AWS service**
Allow AWS services like EC2, Lambda, or others to perform actions in this account.
- ☐ **AWS account**
Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.
- ☐ **Web identity**
Allows users federated by the specified external web identity provider to assume this role to perform actions in this account.
- ☐ **SAML 2.0 federation**
Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.
- ☐ **Custom trust policy**
Create a custom trust policy to enable others to perform actions in this account.

Use case
Allow an AWS service like EC2, Lambda, or others to perform actions in this account.

Service or use case

Elastic Beanstalk

Choose a use case for the specified service.

Use case

- ☐ Elastic Beanstalk - Compute
Allows your environment's EC2 instances to perform operations required for your application.
- ☒ **Elastic Beanstalk - Environment**
Allows access to other AWS service resources that are required to create and manage environments.

Cancel Next

Screenshot: Create EC2 Instance Role

- In EC2 instance profile
- Click Create role
- Select AWS service
- Choose use case as Elastic Beanstalk
- Select Elastic Beanstalk compute
- Click Create role

Step 1: Select trusted entity

Step 2: Add permissions

Step 3: Name, review, and create

Select trusted entity info

Trusted entity type

- ☒ **AWS service**
Allow AWS services like EC2, Lambda, or others to perform actions in this account.
- ☐ **AWS account**
Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.
- ☐ **Web identity**
Allows users federated by the specified external web identity provider to assume this role to perform actions in this account.
- ☐ **SAML 2.0 federation**
Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.
- ☐ **Custom trust policy**
Create a custom trust policy to enable others to perform actions in this account.

Use case
Allow an AWS service like EC2, Lambda, or others to perform actions in this account.

Service or use case

Elastic Beanstalk

Choose a use case for the specified service.

Use case

- ☒ **Elastic Beanstalk - Compute**
Allows your environment's EC2 instances to perform operations required for your application.
- ☐ Elastic Beanstalk - Environment
Allows access to other AWS service resources that are required to create and manage environments.

Cancel Next

Screenshot: Select Roles

- Choose the created service role
- Choose the created EC2 instance role
- Choose key pair

Step 4 - optional

Configure instance traffic and scaling

Step 5 - optional

Configure updates, monitoring, and logging

Step 6

Review

Choose an IAM role for Elastic Beanstalk to assume as a service role. The IAM role must have the required IAM managed policies.

aws-elasticbeanstalk-service-role

Create role

EC2 instance profile

Choose an IAM instance profile with managed policies that allow your EC2 instances to perform required operations.

aws-elasticbeanstalk-ec2-role

Create role

EC2 key pair - optional

Select an EC2 key pair to securely log in to your EC2 instances. [Learn more](#)

3pmbatch2

Cancel

Skip to Review

Previous

Next

Screenshot: Configure Network

- Select default VPC
- Enable public IP
- Select all Availability Zones

Elastic Beanstalk

Create environment

Elastic Beanstalk

Applications

Environments

Change history

Recent environments

Mywebapp-env

Step 1

Configure environment

Step 2

Configure service access

Step 3 - optional

Set up networking, database, and tags

Step 4 - optional

Configure instance traffic and scaling

Step 5 - optional

Configure updates, monitoring, and logging

Step 6

Review

Set up networking, database, and tags - optional

Instance settings

Choose a subnet in each AZ for the instances that run your application. To avoid exposing your instances to the Internet, run your instances in private subnets and load balancer in public subnets. To run your load balancer and instances in the same public subnets, assign public IP addresses to the instances. [Learn more](#)

VPC

Launch your environment in a custom VPC instead of the default VPC. You can create a VPC and subnets in the VPC management console. [Learn more](#)

vpc-00e925f7d875b3505 | (172.31.0.0/16)

Create VPC

Public IP address

Assign a public IP address to the Amazon EC2 instances in your environment.

☒ Enable

Instance subnets

Filter instance subnets

☒	Availability Zone	Subnet	CIDR	Name
☒	ap-south-1c	subnet-00cc3409b9fc886f5	172.31.16.0/20	
☒	ap-south-1b	subnet-02af623bce16f5af0	172.31.0.0/20	
☒	ap-south-1a	subnet-0fb0131d339a7347d	172.31.32.0/20	

Screenshot: Configure Instances

- Select security group
- Select instance type
- Click Create environment

Elastic Beanstalk

Create environment

Elastic Beanstalk

Applications

Environments

Change history

Recent environments

Mywebapp-env

Monitoring interval

5 minute

Instance metadata service (IMDS)

Your environment's platform supports both IMDSv1 and IMDSv2. To enforce IMDSv2, deactivate IMDSv1. [Learn more](#)

IMDSv1

With the current setting, the environment enables only IMDSv2.

☒ Disable

EC2 security groups

Select security groups to control traffic.

EC2 security groups

Choose security groups

sg-02d620c3320d040d | default

default VPC security group

Clear all

Instance types

Add instance types for your environment with your preferred launch order. The order preference only applies to On-Demand Instances and Spot Instances that use the capacity optimized prioritized allocation strategy. We recommend you include at least two instance types. [Learn more](#)

1.

t3.micro

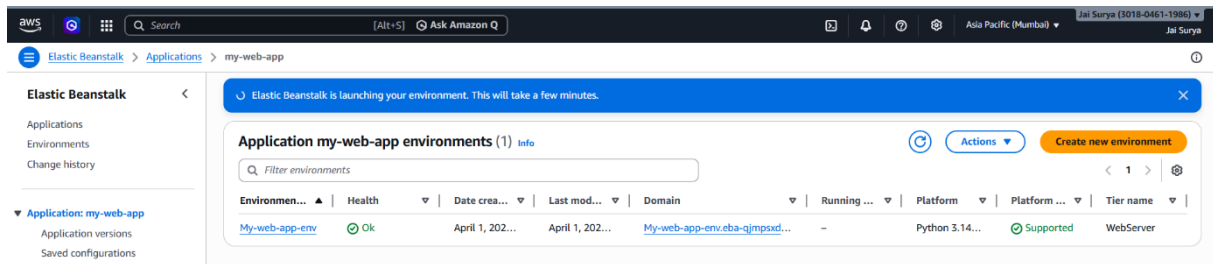
2.

t3.small

Remove

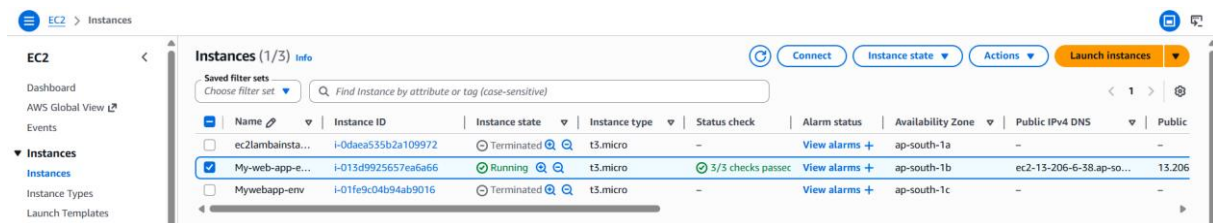
Screenshot: Application Created

- Application created successfully



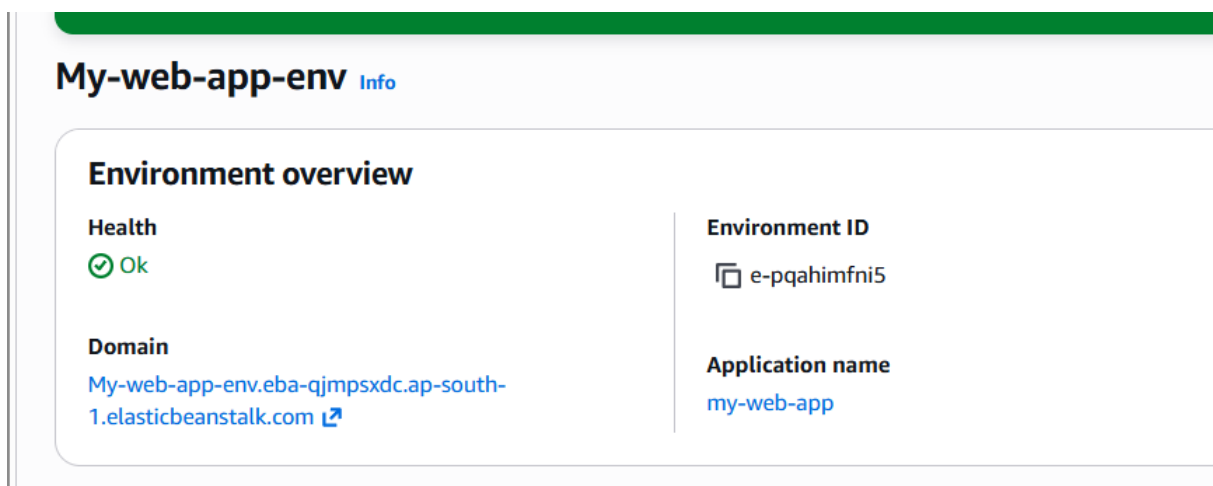
Screenshot: Instances Created

- EC2 instances are created automatically
- Instances are running



Screenshot: Access Application

- Click the application name
- Click the domain name



Screenshot: Application Opened

- Application opens successfully in browser

